Quiz #4
EGRE246

Name:

1. Finish the following function “reverse” to reverse a singly linked list in C, you need to iterate through the list and reverse the direction of each node's `next` pointer (5 points).

```
#include <stdio.h>
#include <stdlib.h>
// Define the structure of a node
struct Node {
   int data;
   struct Node* next;
};
void reverse(struct Node** head_ref);
// add a new node onto the list from the front
void addFront(struct Node** head_ref, int new_data) {
   struct Node* new_node = (struct Node*)malloc(sizeof(struct Node));
   new_node->data = new_data;
   new_node->next = *head_ref;
   *head_ref = new_node;
}
// print the list
void printList(struct Node* node) {
   while (node != NULL) {
      printf("%d -> ", node->data);
      node = node->next;
   }
   printf("NULL\n");
}
// Main function to test the reversal
```

```
int main() {
    struct Node* head = NULL;
    // Create a list 1->2->3->4->5
    addFront (&head, 5);
    addFront (&head, 4);
    addFront (&head, 3);
    addFront (&head, 2);
    addFront (&head, 1);

    printf("Original list: \n");
    printList(head);
    // Reverse the list
    reverse(&head);
    printf("Reversed list: \n");
    printList(head);
    return 0;
}
// Function to reverse the linked list
void reverse(struct Node** head_ref) {
struct Node *cur, *pre, *ptr;
cur = * head_ref;
pre = NULL;
while (cur !=NULL) {
ptr = cur;
ptr->next = pre;
pre = cur;
cur = cur->next
}
*head_ref = pre;
```